Dependency Length, Syntactic Complexity & Memory: A Reading Time Benchmark for Sentence Processing Modeling

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Objective

Create a dataset to evaluate the cognitive plausibility of language models to be used as models of Human Sentence Processing

Context

Expected Human-LLM discrepancy

Dimension	LLMs	Humans
Next-word prediction	Good predictor of surprisal but underpredict human difficulty	Difficulty for unexpected inputs and processing errors
Short-term memory	Decay-free access to huge context window	Lossy access to small context window
Long-term memory	Trained on up to ~15T words	Learn from ~100M words
Generalization efficiency	Often overtrained, relying on memorized surface patterns	Must generalize from sparse, noisy data
Structural processing	Bias toward linear sequence cues; fail to reproduce structure-sensitive errors	Sensitivity to hierarchical structure, syntactic distance, phrase type

Towards cognitively plausible models

Limiting context (e.g., lossy-context surprisal, shorter n-grams, ablated transformers) improves the fit to human reading behavior, supporting the role of memory constraints

Evaluating LMs under human-like memory constraints

- Memory-based theories converge on one point :

Longer dependencies ⇒ harder retrieval ⇒ higher processing cost (locality effects)

- Focus: long dependency resolution as a measure of processing difficulty

Motivation

- **Goal**: obtain a large, controlled RT dataset with increasing dependency length that captures increasing human processing difficulty
- Limitation: large naturalistic reading corpora have low control while psycholinguistic datasets are small

Stimuli structure

- **360 English sentences** (6 conditions × 60 items)
- Systematic **subject–verb dependency length** variation (0 / 4 / 9 words) and **syntactic complexity** (up to 2 embedding depth)
- Generated with **LLaMA-3.2-100B** and **manually reviewed** for semantic coherence, S-V predictability

Biases controlled for:
Antilocality: subject–verb relations kept plausible but not strongly expected • **Priming**: sentences presented in isolation • **Speeding**: all sentences constrained to 13–15 words

Data collection

- **6 experimental lists** (Latin square design)
- **Self-paced reading** (noncumulative moving-window)
- **480 native English speakers** recruited via Prolific
- 50% of sentences followed by **a comprehension questions**
- Working memory capacity measured via an **operation-span task**

	Participants	Sentences
Per list	≈80	60
Total	475	28,500

Clause Type	Dep Length (words)	Example Set
Baseline	0	The violinist leaves the stage before the audience loudly applauds the great performance.
PP	4	The violinist in the large orchestra leaves the stage before the audience loudly applauds.
SRC	4	The violinist that followed the conductor leaves the stage before the audience loudly applauds.
ORC	4	The violinist that the conductor followed leaves the stage before the audience loudly applauds.
2×SRC	9	The violinist that followed the conductor that worked in the orchestra leaves the stage.
2×ORC	9	The violinist that the conductor that the whole orchestra admired followed leaves the stage.

Analysis

- **Models**:
  - Hypothesis testing : linear mixed-effects models (**lmer**)
  - Surprisal extraction: **gpt-2**, **Pythia14m**, **Pythia12b**
- **Normalization**: log-transformed RTs
- Frequencies : **WikiText-103**
- RTs trimming :
  - **<100 ms** or **>3,000 ms** (<1% of data)
  - **≥ 0.7 comprehension accuracy**

Human reading times

Does dependency length predict reading time?

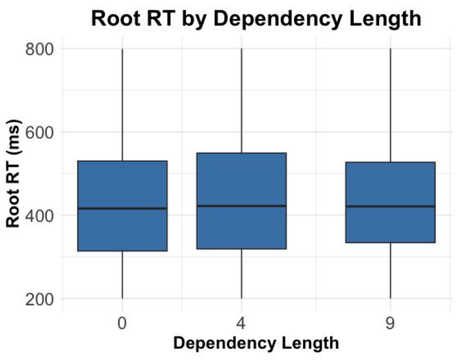
Baseline model: log(root RT) ~ word length + log(freq) + (1 | participant) + (1 | item) + (1 | order)

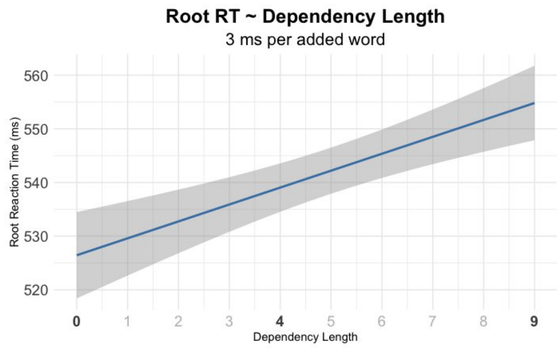
Distance model: log(root RT) ~ **dep length** + word length + log(freq) + (1 | participant) + (1 | item) + (1 | order)

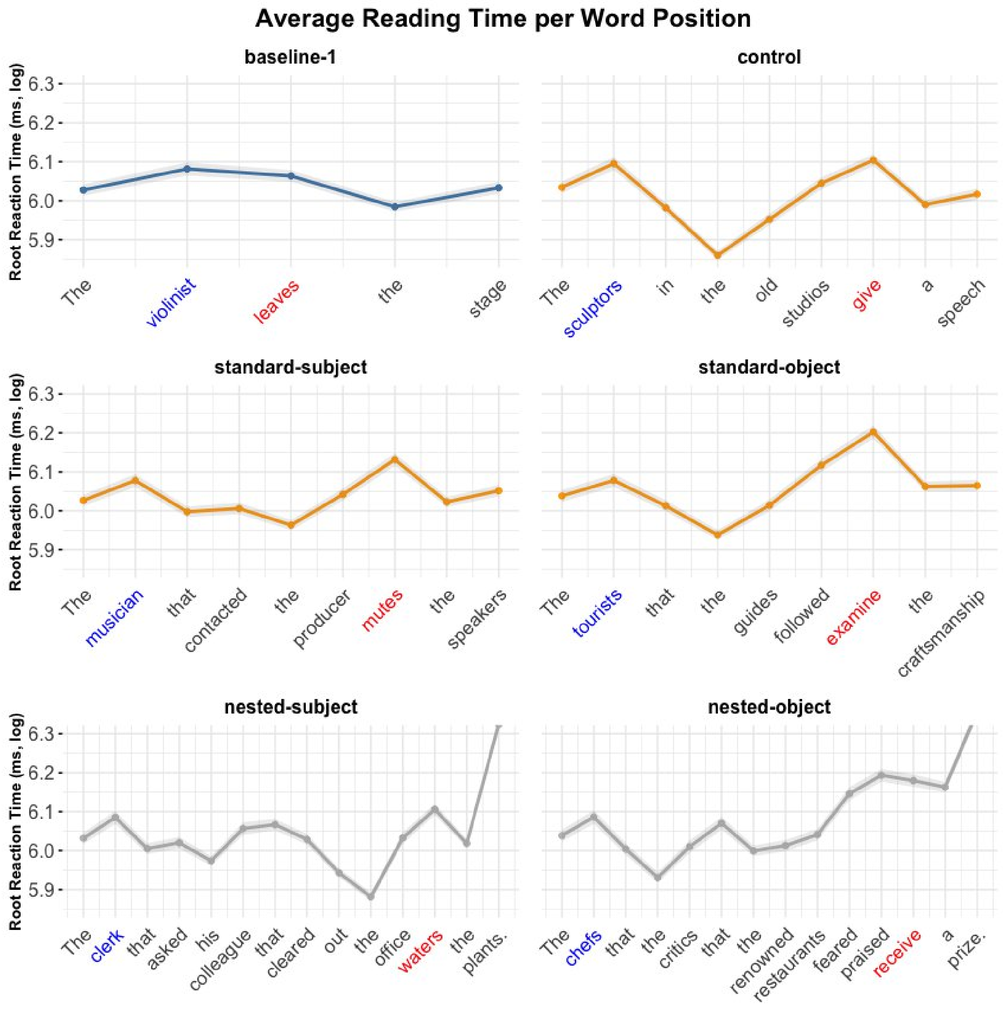
✓ Dependency length significantly improves model fit (χ^2= 21.6, p<0.001)

✓ Longer dependencies lead to longer reading times, independently of word length and frequency

Predictor	Effect on RT	p
Intercept	~430 ms	<0.001
Dep Length	+3 ms per word	<0.001
Word length	+6 ms per character	<0.001
Frequency	−3 ms per log-frequency unit	<0.001

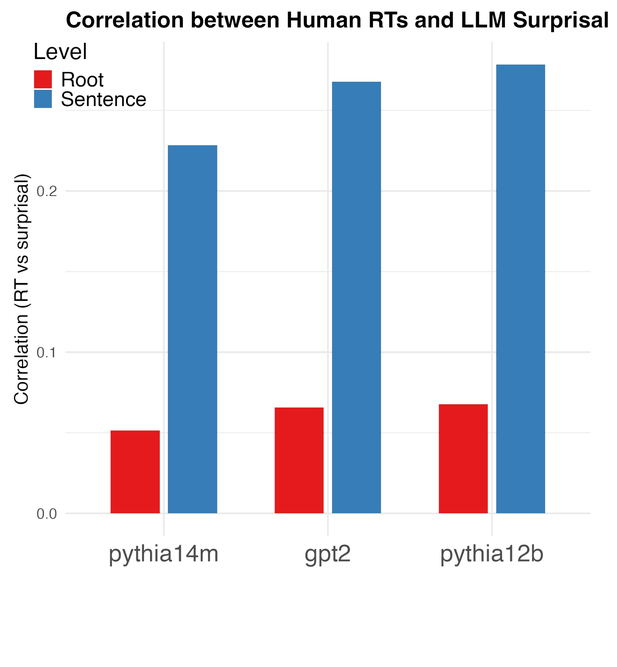




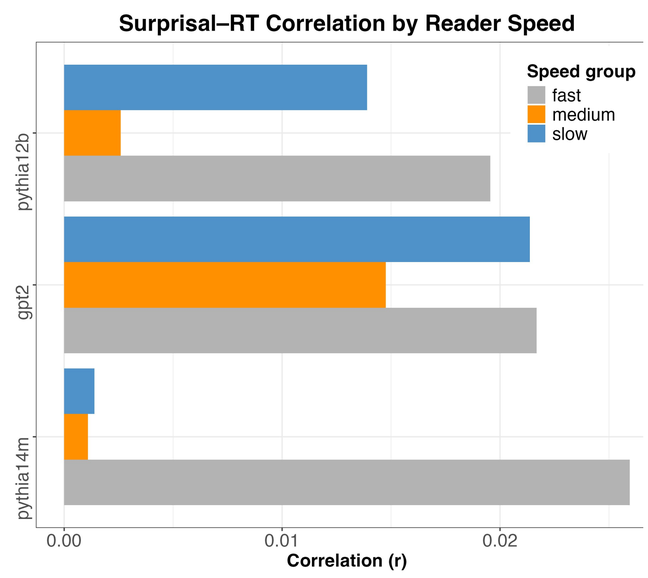


LLM surprisal & human RTs

Which LLM correlates best with RT?



Which reader profile is best predicted?



Contribution

✓ Our dataset replicates **distance effects on dependency resolution**, validating it for testing models of sentence processing

✓ Includes **working-memory measures** for fine-grained modeling of individual variability

✓ LLM **surprisal** tracks **sentence-level variation better** than **word-level RTs at the critical region**; larger models outperform smaller ones ; fast readers are best predicted across all sizes

Acknowledgment

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